

Problem Set #7 — The Variational Equation

University of Colorado - Boulder
CSCI4446 — Chaotic Dynamics

JACK MATHEWS
john.mathewsiii@colorado.edu

March 2, 2026

1 Deriving the Jacobian of the Lorenz System

The Lorenz system is defined as:

$$\vec{F}(\vec{x}, a, r, b) = \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{z} \end{bmatrix} = \begin{bmatrix} a(y - x) \\ rx - y - xz \\ xy - bz \end{bmatrix}$$

The Jacobian matrix in three dimensions is defined as:

$$D_{\vec{x}}\vec{F} = \begin{bmatrix} \frac{\partial f_x}{\partial x} & \frac{\partial f_x}{\partial y} & \frac{\partial f_x}{\partial z} \\ \frac{\partial f_y}{\partial x} & \frac{\partial f_y}{\partial y} & \frac{\partial f_y}{\partial z} \\ \frac{\partial f_z}{\partial x} & \frac{\partial f_z}{\partial y} & \frac{\partial f_z}{\partial z} \end{bmatrix}$$

Applying the Jacobian to the Lorenz system yields:

$$D_{\vec{x}}\vec{F} = \begin{bmatrix} -a & a & 0 \\ r - z & -1 & -x \\ y & x & -b \end{bmatrix}$$

2 Deriving the Variational System

The Variational System $\dot{\delta}$ is defined as the product of the Jacobian and the $n \times n$ matrix of variations δ .

$$\dot{\delta} = D_{\vec{x}}\vec{F}$$

where δ is defined as:

$$\delta = \begin{bmatrix} \delta_{xx} & \delta_{yx} & \delta_{zx} \\ \delta_{xy} & \delta_{yy} & \delta_{zy} \\ \delta_{xz} & \delta_{yz} & \delta_{zz} \end{bmatrix}$$

Thus, for the Lorenz system:

$$\dot{\delta} = \begin{bmatrix} -a & a & 0 \\ r-z & -1 & -x \\ y & x & -b \end{bmatrix} \begin{bmatrix} \delta_{xx} & \delta_{yx} & \delta_{zx} \\ \delta_{xy} & \delta_{yy} & \delta_{zy} \\ \delta_{xz} & \delta_{yz} & \delta_{zz} \end{bmatrix}$$

$$= \begin{bmatrix} -a\delta_{xx} + a\delta_{xy} & -a\delta_{yx} + a\delta_{yy} & -a\delta_{zx} + a\delta_{zy} \\ (r-z)\delta_{xx} - \delta_{xy} - x\delta_{xz} & (r-z)\delta_{yx} - \delta_{yy} - x\delta_{yz} & (r-z)\delta_{zx} - \delta_{zy} - x\delta_{zz} \\ y\delta_{xx} + x\delta_{xy} - b\delta_{xz} & y\delta_{yx} + x\delta_{yy} - b\delta_{yz} & y\delta_{zx} + x\delta_{zy} - b\delta_{zz} \end{bmatrix}$$

3 Simulating the Variational Derivative with RK4

To model variations in the system, we can combine the original system with the variational derivative system into a larger $n^2 + n$ system—called the variational equation—and apply RK4 to this system. That system is written as:

$$\begin{bmatrix} \dot{\vec{x}} \\ \dot{\delta} \end{bmatrix} = \begin{bmatrix} \vec{F} \\ D_{\vec{x}}\vec{F}\delta \end{bmatrix}$$

with the initial conditions:

$$t = t_0, \quad \begin{bmatrix} \vec{x}_0 \\ I \end{bmatrix}$$

4 Implementation and Results

The implementation of my variational equation simulator can be found at line 97 in file `ode_solver/src/systems.cpp` in my Github Repository. The following results are from simulations of the Lorenz attractor using time step $dt = 0.001$ with the parameters $a = 16$, $r = 45$, and $b = 4$ over 100 iterations. All column sums were computed with absolute values—a negative variation doesn't decrease the total variation. (This is equivalent to using a 1-norm to quantify the magnitude of a vector.) Finally, all simulations use the identity matrix as the initial condition of the matrix of variations:

$$\begin{bmatrix} \delta_{xx} & \delta_{xy} & \delta_{xz} \\ \delta_{yx} & \delta_{yy} & \delta_{yz} \\ \delta_{zx} & \delta_{zy} & \delta_{zz} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

4.1 Simulation 1

$$\begin{bmatrix} x_0 & y_0 & z_0 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 2 \end{bmatrix}$$

Final Matrix of Variations:

$$\begin{bmatrix} 2.40394 & 1.91888 & -0.0296751 \\ 5.17512 & 4.20717 & -0.0893006 \\ 0.480783 & 0.374137 & 0.665115 \end{bmatrix}$$

Column Sums:

$$\begin{bmatrix} \sum |\delta_x| & \sum |\delta_y| & \sum |\delta_z| \end{bmatrix} = \begin{bmatrix} 8.05985 & 6.50018 & 0.784091 \end{bmatrix}$$

4.2 Simulation 2

$$\begin{bmatrix} x_0 & y_0 & z_0 \end{bmatrix} = \begin{bmatrix} 10 & -5 & 2 \end{bmatrix}$$

Final Matrix of Variations:

$$\begin{bmatrix} 2.13103 & 1.63916 & -0.485809 \\ 3.86003 & 3.00833 & -1.11058 \\ 3.12194 & 2.53248 & -0.0180607 \end{bmatrix}$$

Column Sums:

$$\begin{bmatrix} \sum |\delta_x| & \sum |\delta_y| & \sum |\delta_z| \end{bmatrix} = \begin{bmatrix} 9.113 & 7.17998 & 1.61445 \end{bmatrix}$$

4.3 Simulation 3

$$\begin{bmatrix} x_0 & y_0 & z_0 \end{bmatrix} = \begin{bmatrix} 0 & -1 & 2 \end{bmatrix}$$

Final Matrix of Variations:

$$\begin{bmatrix} 2.40394 & 1.91888 & 0.0296751 \\ 5.17512 & 4.20717 & 0.0893006 \\ -0.480783 & -0.374137 & 0.665115 \end{bmatrix}$$

Column Sums:

$$\begin{bmatrix} \sum |\delta_x| & \sum |\delta_y| & \sum |\delta_z| \end{bmatrix} = \begin{bmatrix} 8.05985 & 6.50018 & 0.784091 \end{bmatrix}$$

4.4 Analysis

Because this experiment only utilizes three initial conditions on a very short time-scale, there is little to say about the Lorenz system in general, but we can make specific claims about the results with respect to the specific initial conditions. The first thing to note is that perturbations in the x -direction grew the most across all three simulations, suggesting that the system is most sensitive to changes in the x -direction when setting the initial condition around the chosen points. Similarly, the system is least sensitive to perturbations in the z -direction. While the magnitudes of perturbations in x and y grew around seven to nine times their original size, those in z shrank for simulations 1 and 3, and only grew by a factor of 1.6 in simulation 2.

Furthermore, variations from the initial condition in simulation 2 grew the fastest of the three. The largest discrepancies occurred in δ_{xz} and δ_{yz} ; that is, previous z variations had significantly more effect on the x and y variations around the initial condition $[10, -5, 2]$ than around either of the other initial conditions.

Another interesting point to notice is that simulations 1 and 3 produced the same values, although with different signs, *i.e.* the magnitudes were the same. The initial conditions were almost the same—the only difference being the sign in y_0 . This is due to the fact that the Lorenz system is symmetric. It is equivalent across a transformation from $(x, y) \rightarrow (-x, -y)$. This is obvious when plotted with the right perspective, as in Figure 1. Note that the wider trajectories on the right side are from the transient. If enough iterations are eliminated, the system will appear perfectly symmetric (when viewed from above or below).

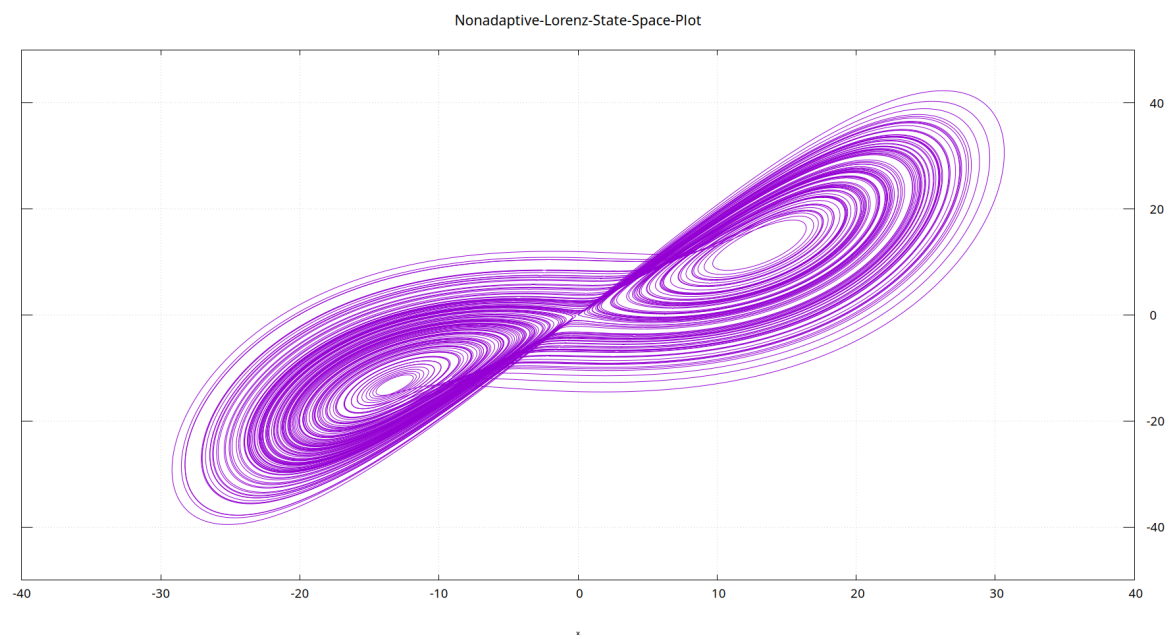


Figure 1: Lorenz Attractor Viewed from Above